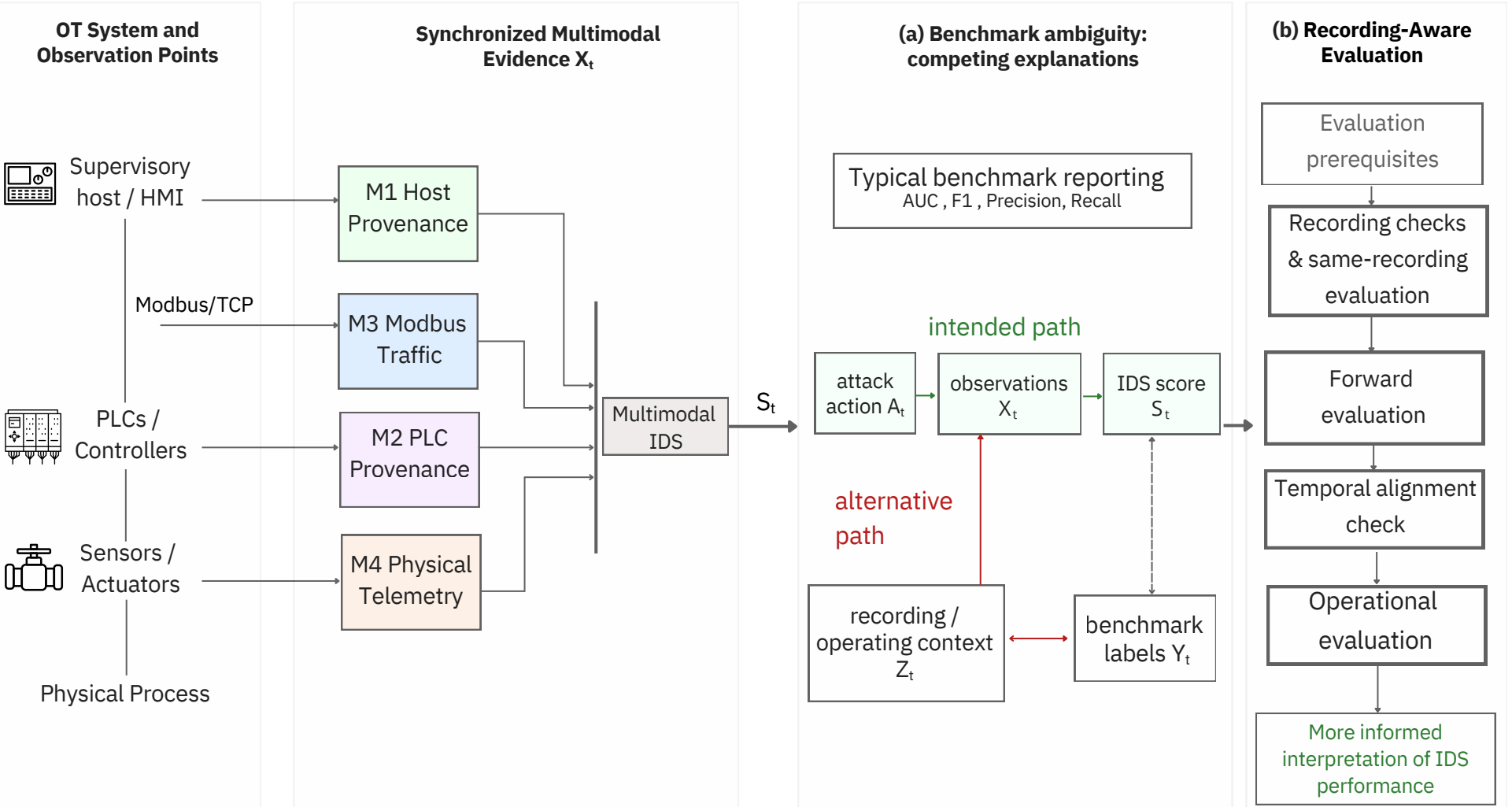

OT System and Observation Points
Supervisory host / HMI
Modbus/TCP
PLCs / Controllers
Sensors / Actuators
Physical Process
Synchronized Multimodal Evidence X_t
M1 Host Provenance
M3 Modbus Traffic
M2 PLC Provenance
M4 Physical Telemetry
Multimodal IDS
S_t
(a) Benchmark ambiguity: competing explanations
Typical benchmark reporting
AUC , F1 , Precision, Recall
intended path
attack action A_t
observations X_t
IDS score S_t
alternative path
recording / operating context Z_t
benchmark labels Y_t
(b) Recording-Aware Evaluation
Evaluation prerequisites
Recording checks & same-recording evaluation
Forward evaluation
Temporal alignment check
Operational evaluation
More informed interpretation of IDS performance